

MATHLETICS

Trigonometry

Student Book - Series J-1

$\sin \theta$ $\cos \theta$
 $\tan \theta$



Mathletics
Instant
Workbooks



Trigonometry

Student Book - Series J

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Practice Tests

Topic 1 - Topic test A

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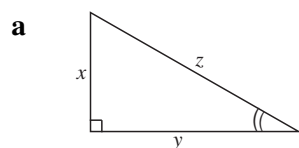
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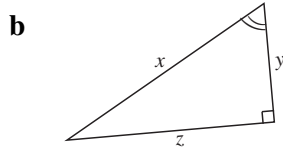
Author of The Topics and Topic Tests: AS Kalra

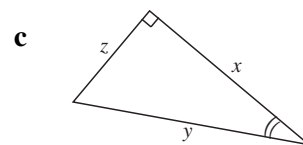
Trigonometry

Topic 1: Naming the sides of a right-angled triangle

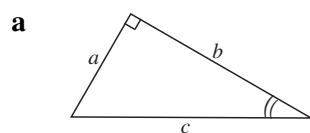
QUESTION 1 In each of the following triangles, state whether x , y and z are the opposite side, adjacent side or hypotenuse, with reference to the angle marked.

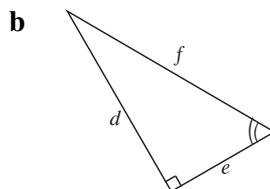


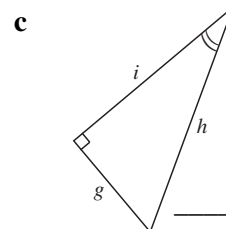


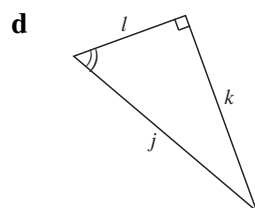


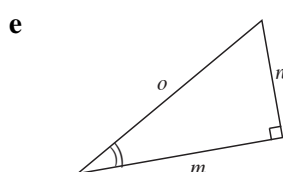
QUESTION 2 Name each side of the following triangles as opposite (opp), adjacent (adj) or hypotenuse (hyp), with reference to the angle marked.

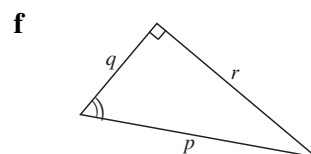




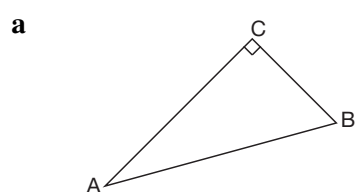


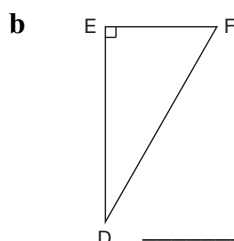


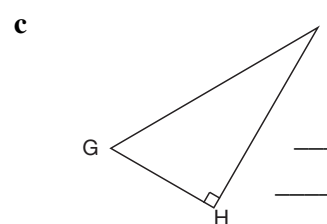


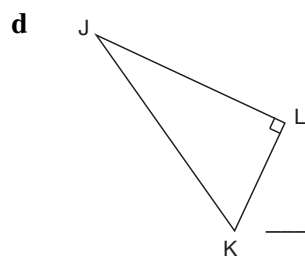


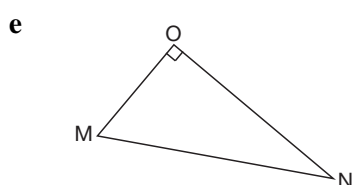
QUESTION 3 Name the hypotenuse in each triangle.

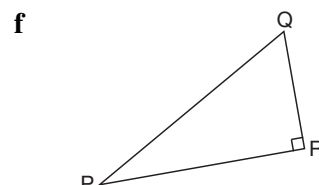








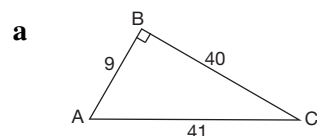




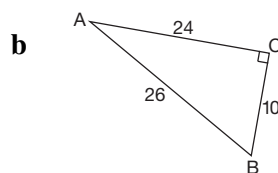
Trigonometry

Topic 2: The sine ratio

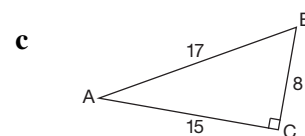
QUESTION 1 Find the value of $\sin A$ in each triangle as a fraction in its simplest form. All lengths are in millimetres.



$$\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{BC}{AC} = \frac{40}{41}$$

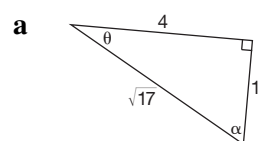


$$\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{BC}{AB} = \frac{10}{26} = \frac{5}{13}$$



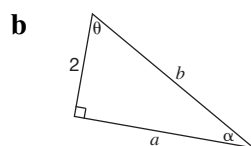
$$\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{BC}{AB} = \frac{8}{17}$$

QUESTION 2 For each triangle, find $\sin \theta$ and $\sin \alpha$. All lengths are in millimetres.



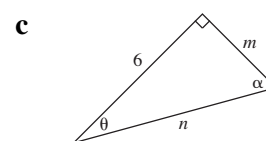
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{1}{\sqrt{17}}$$

$$\sin \alpha = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{4}{\sqrt{17}}$$



$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{a}{b}$$

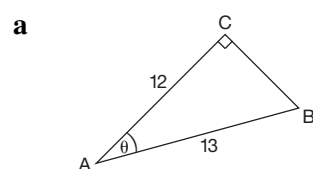
$$\sin \alpha = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{2}{b}$$



$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{n}{m}$$

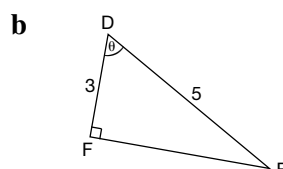
$$\sin \alpha = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{6}{m}$$

QUESTION 3 Use Pythagoras' theorem to calculate the unknown side of each triangle and then find $\sin \theta$.



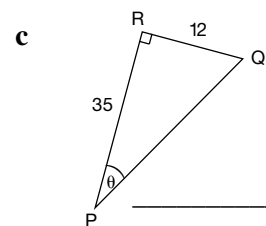
$$\text{Using Pythagoras' theorem: } BC^2 = AB^2 - AC^2 = 13^2 - 12^2 = 25 \Rightarrow BC = 5$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{BC}{AB} = \frac{5}{13}$$



$$\text{Using Pythagoras' theorem: } FE^2 = DE^2 - DF^2 = 5^2 - 3^2 = 16 \Rightarrow FE = 4$$

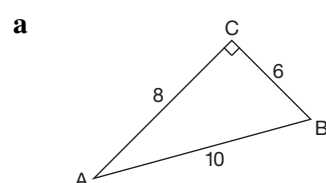
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{FE}{DE} = \frac{4}{5}$$



$$\text{Using Pythagoras' theorem: } QR^2 = PQ^2 - PR^2 = 35^2 - 12^2 = 1159 \Rightarrow QR = \sqrt{1159}$$

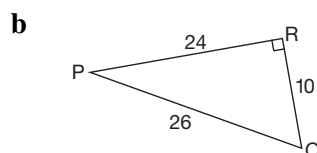
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{QR}{PQ} = \frac{\sqrt{1159}}{35}$$

QUESTION 4 Write each sine ratio in its simplest form.



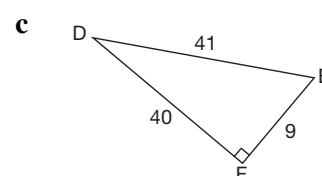
$$\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{BC}{AB} = \frac{6}{10} = \frac{3}{5}$$

$$\sin B = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{AC}{AB} = \frac{8}{10} = \frac{4}{5}$$



$$\sin P = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{QR}{PQ} = \frac{10}{26} = \frac{5}{13}$$

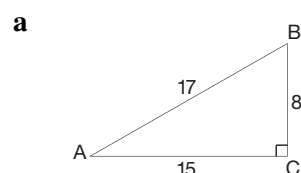
$$\sin Q = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{PR}{PQ} = \frac{24}{26} = \frac{12}{13}$$



$$\sin D = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{EF}{DE} = \frac{9}{40}$$

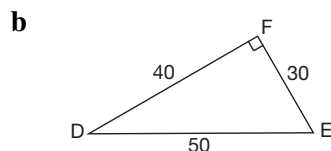
$$\sin E = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{DF}{DE} = \frac{41}{40}$$

QUESTION 5 Name the angle of each triangle that has the given sine ratio.



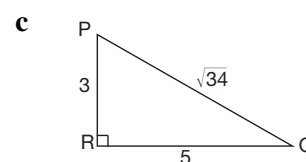
$$\sin \square = \frac{8}{17} \Rightarrow \text{Angle } B$$

$$\sin \square = \frac{15}{17} \Rightarrow \text{Angle } A$$



$$\sin \square = \frac{30}{50} = \frac{3}{5} \Rightarrow \text{Angle } E$$

$$\sin \square = \frac{40}{50} = \frac{4}{5} \Rightarrow \text{Angle } D$$



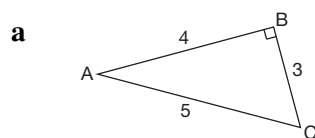
$$\sin \square = \frac{5}{\sqrt{34}} \Rightarrow \text{Angle } P$$

$$\sin \square = \frac{3}{\sqrt{34}} \Rightarrow \text{Angle } Q$$

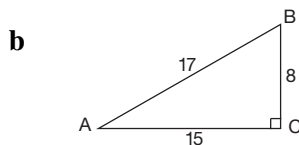
Trigonometry

Topic 3: The cosine ratio

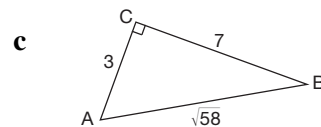
QUESTION 1 Find the value of $\cos A$ in each triangle as a fraction in its simplest form. All lengths are in millimetres.



$$\cos A = \underline{\hspace{2cm}}$$

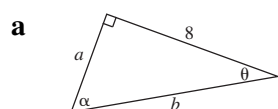


$$\cos A = \underline{\hspace{2cm}}$$



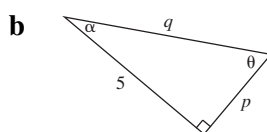
$$\cos A = \underline{\hspace{2cm}}$$

QUESTION 2 For each triangle, find $\cos \theta$ and $\cos \alpha$. All lengths are in millimetres.



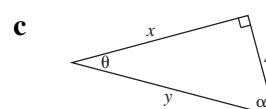
$$\cos \theta = \underline{\hspace{2cm}}$$

$$\cos \alpha = \underline{\hspace{2cm}}$$



$$\cos \theta = \underline{\hspace{2cm}}$$

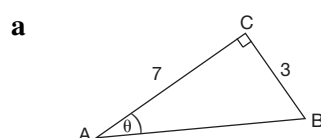
$$\cos \alpha = \underline{\hspace{2cm}}$$



$$\cos \theta = \underline{\hspace{2cm}}$$

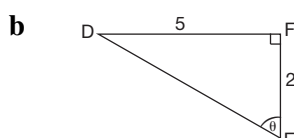
$$\cos \alpha = \underline{\hspace{2cm}}$$

QUESTION 3 Use Pythagoras' theorem to calculate the unknown side and then find $\cos \theta$.



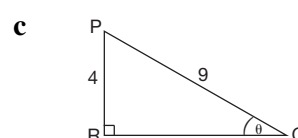
$$\underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}}$$



$$\underline{\hspace{2cm}}$$

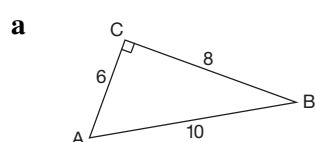
$$\underline{\hspace{2cm}}$$



$$\underline{\hspace{2cm}}$$

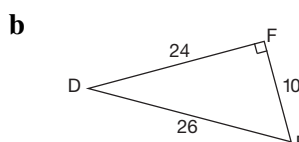
$$\underline{\hspace{2cm}}$$

QUESTION 4 Write each cosine ratio in its simplest form.



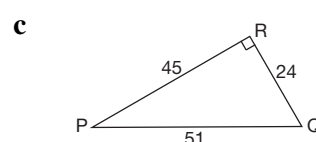
$$\cos A = \underline{\hspace{2cm}}$$

$$\cos B = \underline{\hspace{2cm}}$$



$$\cos D = \underline{\hspace{2cm}}$$

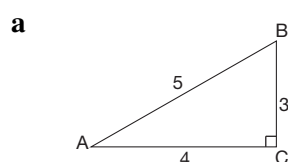
$$\cos E = \underline{\hspace{2cm}}$$



$$\cos P = \underline{\hspace{2cm}}$$

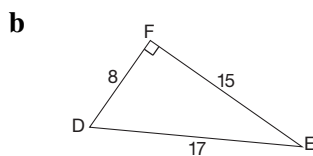
$$\cos Q = \underline{\hspace{2cm}}$$

QUESTION 5 Name the angle of each triangle that has the given cosine ratio.



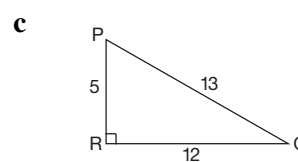
$$\cos \square = \frac{4}{5} \underline{\hspace{2cm}}$$

$$\cos \square = \frac{3}{5} \underline{\hspace{2cm}}$$



$$\cos \square = \frac{8}{17} \underline{\hspace{2cm}}$$

$$\cos \square = \frac{15}{17} \underline{\hspace{2cm}}$$



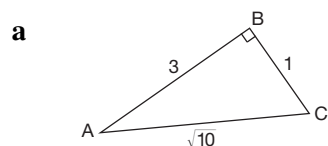
$$\cos \square = \frac{5}{13} \underline{\hspace{2cm}}$$

$$\cos \square = \frac{12}{13} \underline{\hspace{2cm}}$$

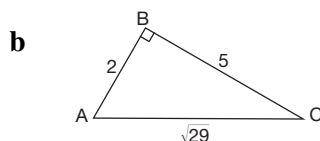
Trigonometry

Topic 4: The tangent ratio

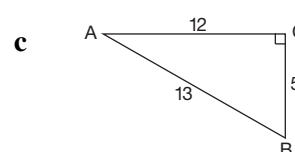
QUESTION 1 Find the value of $\tan A$ in each triangle as a fraction in its simplest form. All lengths are in millimetres.



$\tan A =$ _____

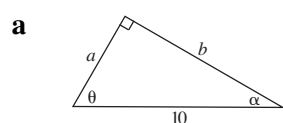


$\tan A =$ _____



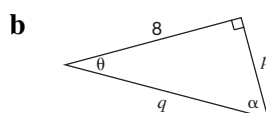
$\tan A =$ _____

QUESTION 2 For each triangle, find $\tan \theta$ and $\tan \alpha$. All lengths are in millimetres.



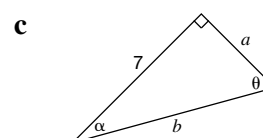
$\tan \theta =$ _____

$\tan \alpha =$ _____



$\tan \theta =$ _____

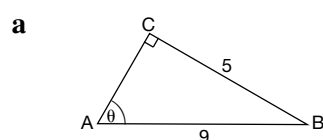
$\tan \alpha =$ _____

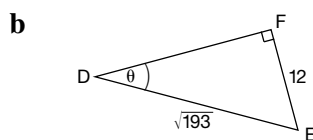


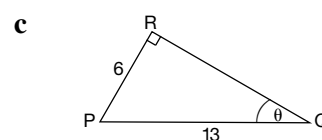
$\tan \theta =$ _____

$\tan \alpha =$ _____

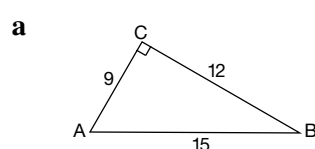
QUESTION 3 Use Pythagoras' theorem to find the unknown side and then find $\tan \theta$.





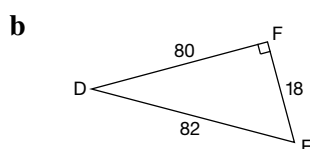


QUESTION 4 Write each tangent ratio in its simplest form.



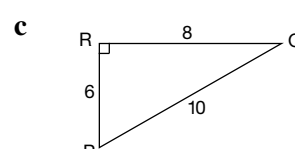
$\tan A =$ _____

$\tan B =$ _____



$\tan D =$ _____

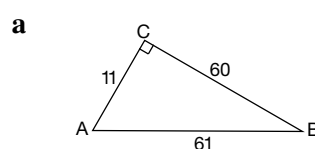
$\tan E =$ _____



$\tan P =$ _____

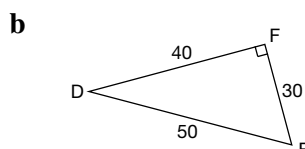
$\tan Q =$ _____

QUESTION 5 Name the angle of each triangle that has the given tangent ratio.



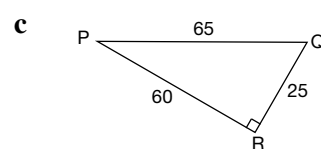
$\tan \square = \frac{60}{11}$ _____

$\tan \square = \frac{11}{60}$ _____



$\tan \square = \frac{30}{40}$ _____

$\tan \square = \frac{40}{30}$ _____



$\tan \square = \frac{25}{60}$ _____

$\tan \square = \frac{60}{25}$ _____

Trigonometry

Topic 5: Use of a calculator in trigonometry

QUESTION 1 Find the value of the following correct to two decimal places.

- a $\sin 34^\circ =$ _____ b $\tan 70^\circ =$ _____ c $\cos 15^\circ =$ _____
d $\cos 59^\circ =$ _____ e $\cos 40^\circ =$ _____ f $\sin 38^\circ =$ _____
g $\tan 83^\circ =$ _____ h $\sin 30^\circ =$ _____ i $\tan 64^\circ =$ _____

QUESTION 2 Find the value of the following correct to three decimal places.

- a $\frac{\sin 35^\circ}{2} =$ _____ b $\frac{\cos 64^\circ}{8} =$ _____ c $\frac{18.9}{\cos 35^\circ} =$ _____
d $\frac{\cos 38^\circ 42'}{2.5} =$ _____ e $\frac{\sin 29^\circ 43'}{8.4} =$ _____ f $\frac{20.5}{\sin 53^\circ 27'} =$ _____
g $\frac{\tan 29^\circ 18'}{7.25} =$ _____ h $\frac{\tan 68^\circ 25'}{7.1} =$ _____ i $\frac{829}{\tan 28^\circ 15'} =$ _____

QUESTION 3 Find the value of the following correct to three significant figures.

- a $3.9 \tan 23^\circ =$ _____ b $\tan 56^\circ 8' =$ _____ c $\cos 35^\circ 29' =$ _____
d $7 \sin 35^\circ =$ _____ e $\sin 25^\circ 19' =$ _____ f $\sin 69^\circ 18' =$ _____
g $\cos 61^\circ 38' =$ _____ h $8.4 \cos 65^\circ 23' =$ _____ i $\tan 23^\circ 46' =$ _____

QUESTION 4 A is an acute angle. Find its size to the nearest degree.

- a $\sin A = 0.6325$ _____ b $\cos A = 0.3787$ _____ c $\tan A = 2.538$ _____
d $\cos A = 0.5783$ _____ e $\tan A = 0.7938$ _____ f $\sin A = 0.7613$ _____
g $\tan A = 1.6928$ _____ h $\sin A = 0.2831$ _____ i $\cos A = 0.9852$ _____

QUESTION 5 A is an acute angle. Find its size to the nearest degree.

- a $\sin A = 0.5$ _____ b $\tan A = 0.5832$ _____ c $\sin A = 0.7681$ _____
d $\cos A = 0.3876$ _____ e $\cos A = 0.5$ _____ f $\tan A = 2.1075$ _____

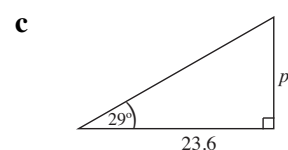
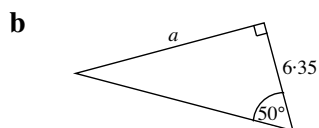
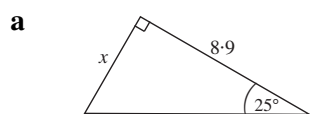
QUESTION 6 Find the size of the acute angle B to the nearest degree.

- a $\tan B = \frac{16}{23}$ _____ b $\cos B = \frac{5}{13}$ _____ c $\sin B = \frac{8.3}{14.5}$ _____
d $\sin B = \frac{1}{2}$ _____ e $\tan B = \frac{8}{9}$ _____ f $\cos B = \frac{11.3}{14.8}$ _____

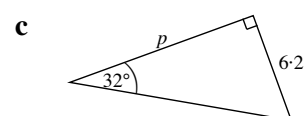
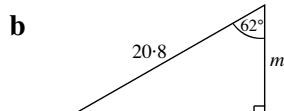
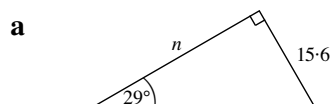
Trigonometry

Topic 6: Finding an unknown side

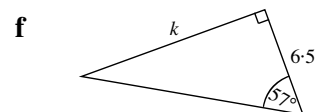
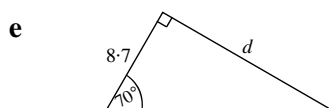
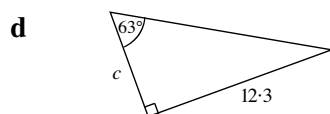
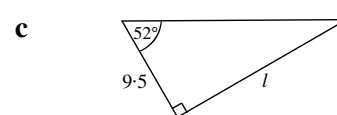
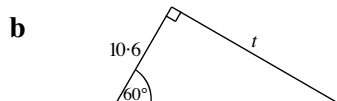
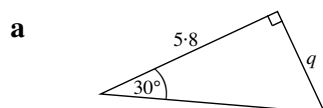
QUESTION 1 Find the value of the unknown side correct to one decimal place.



QUESTION 2 Find the value of the pronumeral in the following triangles correct to two decimal places.



QUESTION 3 Find the value of the pronumeral correct to two decimal places.



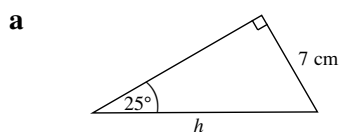
QUESTION 4 A piece of wood 2.5 m long leans against a vertical wall, making an angle of 51° with the floor. How far up the wall, to the nearest centimetre, is the top of the wooden piece?

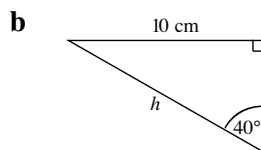
QUESTION 5 In $\triangle PQR$, $\angle R = 90^\circ$, $\angle P = 48^\circ$ and $PQ = 8.6$ cm, find PR correct to two decimal places.

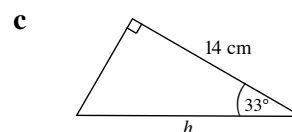
Trigonometry

Topic 7: Finding the hypotenuse

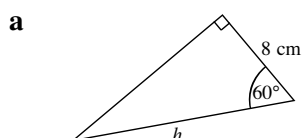
QUESTION 1 Find the length of the hypotenuse correct to one decimal place.

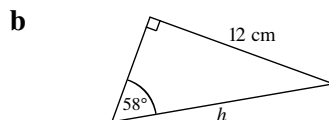


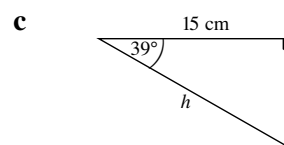




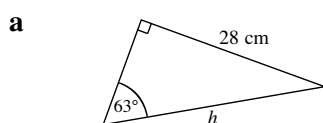
QUESTION 2 Find the length of the hypotenuse correct to two decimal places.

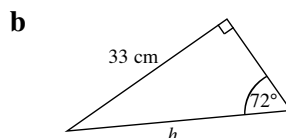


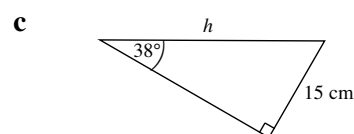


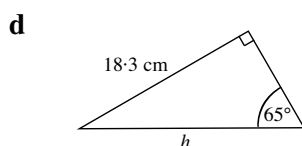


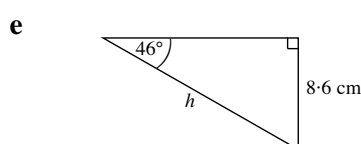
QUESTION 3 Find the length of the hypotenuse correct to one decimal place.

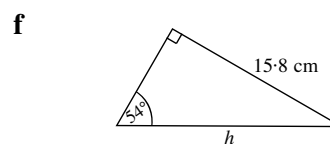




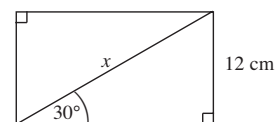








QUESTION 4 Find the value of x .

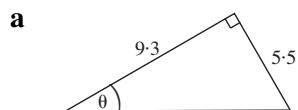


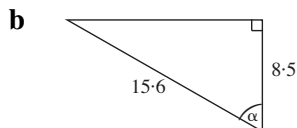
QUESTION 5 In $\triangle ABC$, $\angle A = 90^\circ$, $\angle B = 58^\circ$ and $AB = 23$ m, find BC correct to the nearest metre.

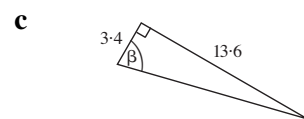
Trigonometry

Topic 8: Finding an unknown angle

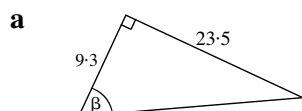
QUESTION 1 Find the size of the angle marked with pronumeral to the nearest degree.

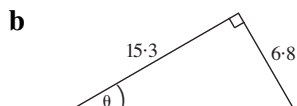


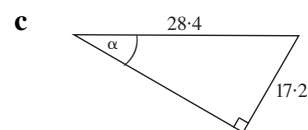




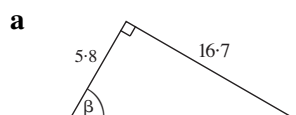
QUESTION 2 Find the size of the angle marked to the nearest degree.

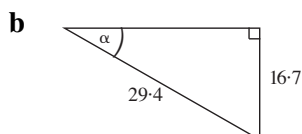


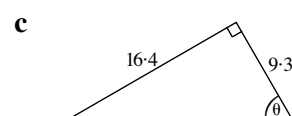


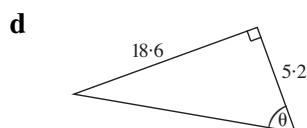


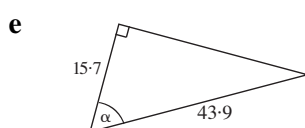
QUESTION 3 Find the size of the angle marked to the nearest degree.

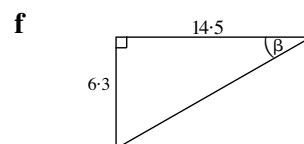






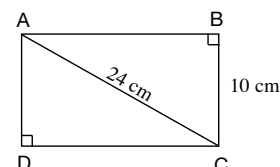






QUESTION 4 In $\triangle PQR$, $\angle R = 90^\circ$, $QR = 8.2$ cm and $PR = 6.7$ cm, find $\angle P$ to the nearest degree.

QUESTION 5 ABCD is a rectangle with $AC = 24$ cm and $AD = 10$ cm. Find $\angle ACD$ correct to the nearest degree.

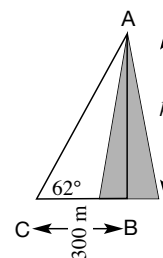


Trigonometry

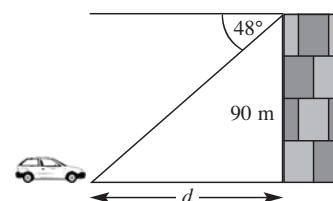
Topic 9: Angles of elevation and depression

QUESTION 1

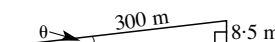
- a** The angle of elevation of the top of a tower AB is 62° from a point C on the ground 300 m from the middle of the base of the tower. Calculate the height of the tower to the nearest metre. _____



- b** From the top of a building 90 m tall, the angle of depression of a car parked on the ground is 48° . Find the distance of the car from the base of the building. Write your answer correct to two decimal places. _____

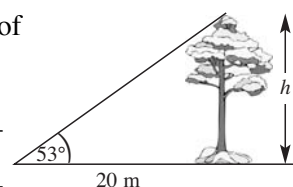


- c** A railway track rises uniformly 8.5 m for every 300 m along the track. Find the angle of elevation of this track to the nearest degree. _____

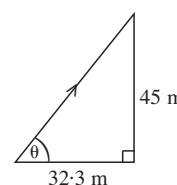


QUESTION 2

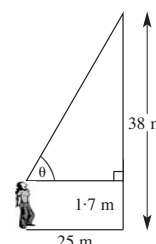
- a** From a point on the ground 20 m from the base of a tree, the angle of elevation of the top of the tree is 53° . Find the height of the tree to the nearest metre. _____



- b** A building that is 45 m tall casts a horizontal shadow 32.3 m long. Find the angle of elevation of the sun to the nearest degree. _____



- c** Anna is 1.70 m tall and is 25 metres away from a building 38 m high. What is the angle of elevation of the top of the building from her eyes? Answer to the nearest degree. _____



Trigonometry

Topic Test

PART A

Time allowed: 15 minutes

Total marks = 15

		Marks
1	Use your calculator to find $\cos 48^\circ$ correct to two decimal places. (A) 0.74 (B) 1.11 (C) 0.67 (D) none of these	1
2	Evaluate $25 \tan 63^\circ$ correct to two decimal places. (A) 1.96 (B) 49.07 (C) 29.38 (D) 22.28	1
3	Find the value of $\frac{\cos 32^\circ}{43.27}$ correct to two decimal places. (A) 0.01 (B) 0.02 (C) 0.03 (D) 0.0196	1
4	If $\sin \theta = \frac{5}{9}$, calculate the size of angle θ to the nearest degree. (A) 31° (B) 32° (C) 33° (D) 34°	1
5	The hypotenuse of a right-angled triangle is 41 cm. If one side is 40 cm, the third side is (A) 1 cm (B) 9 cm (C) 10 cm (D) 81 cm	1
6	If $\cos \theta = \frac{1}{2}$, find the size of angle θ . (A) 30° (B) 45° (C) 60° (D) 72°	1
7	28.65° equals (A) $29^\circ 5'$ (B) $28^\circ 39'$ (C) $29^\circ 39'$ (D) $28^\circ 5'$	1
8	The three sides of a right-angled triangle measure 15 cm, 17 cm and 8 cm. The length of the hypotenuse is (A) 8 cm (B) 15 cm (C) 17 cm (D) 25 cm	1
9	Find the size of angle θ to the nearest degree. (A) 40° (B) 41° (C) 42° (D) 58°	1
10	In a $\triangle ABC$, the angle B is 90° , AB is 8 m and AC is 10 m. Find the size of angle A correct to the nearest degree. (A) 36° (B) 39° (C) 53° (D) 37°	1
11	A road rises uniformly 30.6 m for every 600 m along the road. Find the angle of elevation of this road correct to the nearest degree. (A) 1° (B) 2° (C) 3° (D) 4°	1
12	Find the hypotenuse of this triangle in centimetres correct to 1 decimal place. (A) 9 cm (B) 15.1 cm (C) 12.8 cm (D) none of these	1
13	Use your calculator to find $7.9 \cos 63^\circ$ correct to three significant figures. (A) 3.58 (B) 3.59 (C) 7.03 (D) 7.04	1
14	Evaluate $\frac{\sin 54^\circ}{28.65}$ correct to two decimal places. (A) 0.02 (B) 0.03 (C) 0.04 (D) 0.05	1
15	Find the size of the acute angle θ to the nearest degree if $\tan \theta = \frac{12.5}{19.34}$ (A) 40° (B) 32° (C) 33° (D) none of these	1

Total marks achieved for PART A

15

Trigonometry

Topic Test

PART B

Time allowed: 15 minutes

Total marks = 15

Marks

Question 1

a Find the value of each expression correct to two decimal places.

i $\frac{\cos 72^\circ}{8.93} =$ _____

ii $\frac{72.54}{\tan 68^\circ} =$ _____

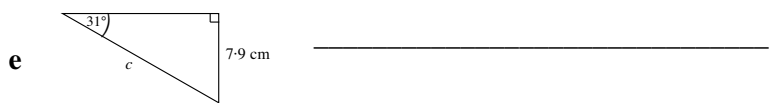
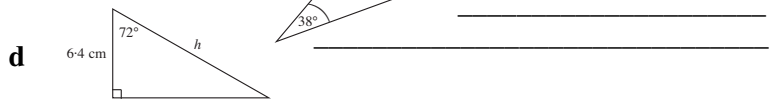
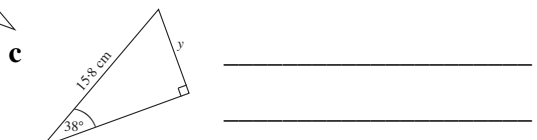
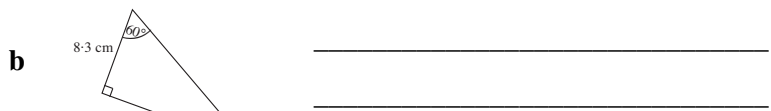
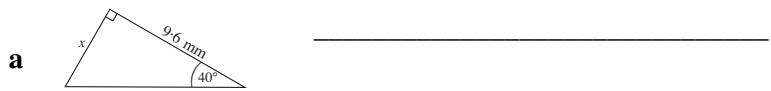
iii $\frac{34.20}{\sin 56^\circ} =$ _____

b Find acute angle A to the nearest degree.

i $\sin A = 0.6835$ _____

ii $\tan A = 1.4862$ _____

Question 2 Find the value of the pronumeral in each triangle correct to two decimal places.



Question 3 The diagonal of a square is 12.5 cm long.

a Find the length of one side correct to the nearest mm.

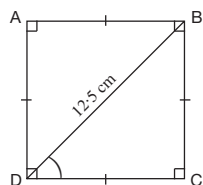
b Find the size of $\angle BDC$.

Find the following correct to three decimal places.

c $\sin \angle BDC$

d $\cos \angle DBC$

e $\tan \angle ABD$



Total marks achieved for PART B

15